

Solutions to Supp/Deferred Examination

1.(a) We are assuming that there is at least one integer $k \geq 0$ such that $a_k \neq b_k$. According to the Well-Ordering Principle, there must be a smallest integer k with this property.

(b) If $a_k = b_k$ for all integers k with $0 \leq k < n$, then

$$\begin{aligned} \sum_{k=0}^{\infty} (a_k - b_k)(z - z_0)^k &= \sum_{k=n}^{\infty} (a_k - b_k)(z - z_0)^k \\ &= \sum_{k=0}^{\infty} (a_{n+k} - b_{n+k})(z - z_0)^{n+k} \\ &= (z - z_0)^n \sum_{k=0}^{\infty} (a_{n+k} - b_{n+k})(z - z_0)^k. \end{aligned}$$

(c) Since $\sum_{k=0}^{\infty} a_k(z - z_0)^k = f(z) = \sum_{k=0}^{\infty} b_k(z - z_0)^k$, we have that

$$\sum_{k=0}^{\infty} (a_k - b_k)(z - z_0)^k = 0$$

for all $z \in D_R$. According to equation (*), it follows that

$$(z - z_0)^n \sum_{k=0}^{\infty} (a_{n+k} - b_{n+k})(z - z_0)^k = 0.$$

If $z \neq z_0$, we can divide by $(z - z_0)^n$ to get the required result.

(d) If $z = z_0$, then

$$\sum_{k=0}^{\infty} (a_{n+k} - b_{n+k})(z - z_0)^k = a_n - b_n \neq 0.$$

(e) (**) says that $\sum_{k=0}^{\infty} (a_{n+k} - b_{n+k})(z - z_0)^k$ converges on D_R to a function which is non-zero at $z = z_0$. Since this function is continuous at $z = z_0$, it must be non-zero on a disc about z_0 . But in part (c) we showed that this function is zero for all $z \neq z_0$ in D_R .

2.(a) Use the Cauchy-Hadamard Formula:

$$\frac{1}{R} = \limsup_{k \in \mathbb{N}} \left(\frac{k + (-1)^k}{k} \right)^k = \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k} \right)^k = e.$$

Thus $R = \frac{1}{e}$.

(b) Use the Ratio Formula:

$$R = \lim_{k \rightarrow \infty} \frac{(k!)^2 (2k+2)!}{(2k)! ((k+1)!)^2} = \lim_{k \rightarrow \infty} \frac{(2k+1)(2k+2)}{(k+1)^2} = 4.$$

3

3.(a) We split the path into two parts. The line segment joining 2 and -2 can be parametrised by $\gamma_1(x) = -x$, where $-2 \leq x \leq 2$, so

$$\int_{\gamma_1} |z|^2 dz = - \int_{-2}^2 x^2 dx = -\frac{16}{3}.$$

The semicircle can be parametrised by $\gamma_2(t) = -2e^{it}$, where $0 \leq t \leq \pi$. This yields

$$\int_{\gamma_2} |z|^2 dz = - \int_0^\pi 8ie^{it} dt = 16.$$

Thus $\int_{\gamma} |z|^2 dz = \frac{32}{3}.$

6

(b) The only singularity is a simple pole at $z = -\frac{1}{2}$, which is inside the curve traced out by γ .

$$\int_{\gamma} \frac{ze^z}{2z+1} dz = 2\pi i \operatorname{Res}_{z=-1/2} \frac{ze^z}{2z+1} = 2\pi i \frac{-\frac{1}{2}e^{-1/2}}{2} = -\frac{1}{2}\pi i e^{-1/2}.$$

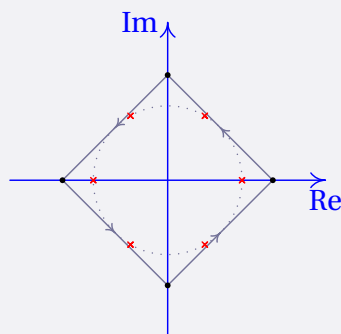
4

(c) The only singularity inside the curve is a double pole at $z = 0$, so

$$\begin{aligned} \int_{\gamma} \frac{1}{z \sin(z)} dz &= 2\pi i \operatorname{Res}_{z=0} \frac{1}{z \sin(z)} \\ &= 2\pi i \lim_{z \rightarrow 0} \frac{d}{dz} \left(\frac{z}{\sin(z)} \right) \\ &= 2\pi i \lim_{z \rightarrow 0} \frac{\sin(z) - z \cos(z)}{\sin^2(z)} \\ &= 2\pi i \lim_{z \rightarrow 0} \frac{(z - \frac{1}{6}z^3) - z(z - \frac{1}{2}z^2)}{z^2} \\ &= 2\pi i \lim_{z \rightarrow 0} \frac{\frac{1}{3}z^3}{z^2} \\ &= 0. \end{aligned}$$

5

(d) The integrand has six simple poles that all lie inside the curve traced out by γ .



We can use the residue at infinity:

$$\begin{aligned}\int_{\gamma} \frac{z^5}{8z^6 - 1} dz &= -2\pi i \operatorname{Res}_{z=\infty} \frac{z^5}{8z^6 - 1} \\ &= 2\pi i \operatorname{Res}_{w=0} \frac{w^{-5}}{w^2(8w^{-6} - 1)} \\ &= 2\pi i \operatorname{Res}_{w=0} \frac{1}{w(8 - w^6)} \\ &= \frac{\pi}{4} i.\end{aligned}$$

5

(The argument principle also works.)

4.(a) First note that

$$\int_0^{\infty} \frac{\cos x}{(9x^2 + 1)^2} dx = \frac{1}{2} \int_{-\infty}^{\infty} \frac{e^{ix}}{(9x^2 + 1)^2} dx.$$

There are no problems concerning convergence. The function $f(z) = \frac{e^{iz}}{(9z^2 + 1)^2}$ has a double pole at $\frac{1}{3}i$. Put $\varphi(z) = f(z)(z - \frac{1}{3}i)^2 = \frac{e^{iz}}{81(z + \frac{1}{3}i)^2}$. Then

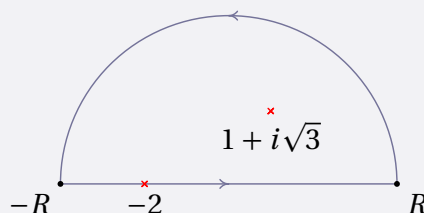
$$\begin{aligned}\operatorname{Res}_{z=i/3} \frac{e^{iz}}{(9z^2 + 1)^2} &= \varphi'(\tfrac{1}{3}i) = \frac{ie^{-1/3}(\tfrac{2}{3}i)^2 - e^{-1/3}2(\tfrac{2}{3}i)}{81(\tfrac{2}{3}i)^4} \\ &= \frac{i(\tfrac{2}{3}i) - 2}{81(\tfrac{2}{3}i)^3 e^{1/3}} \\ &= \frac{1}{9ie^{1/3}}.\end{aligned}$$

Thus

$$\int_0^{\infty} \frac{\cos x}{(9x^2 + 1)^2} dx = \frac{1}{2} \cdot 2\pi i \cdot \frac{1}{9ie^{1/3}} = \frac{\pi}{9e^{1/3}}.$$

8

(b) The function $f(z) = \frac{z^2 + 2}{z^3 + 8}$ has three simple poles, at -2 and at $1 \pm i\sqrt{3}$. Let $\mathcal{C}$ be the curve made up of the segment of the real axis from $-R$ to R and the semicircle traced out by $\gamma(t) = Re^{it}$, where $0 \leq t \leq \pi$. If $R > 2$, then this curve will pass through the pole at -2 and enclose the pole at $1 + i\sqrt{3}$.



We have that

$$\begin{aligned}\int_{\mathcal{C}} f(z) dz &= 2\pi i \left(\frac{1}{2} \operatorname{Res}_{z=-2} f(z) + \operatorname{Res}_{z=1+i\sqrt{3}} f(z) \right) \\ &= 2\pi i \left(\frac{1}{2} \cdot \frac{4+2}{3 \times 4} + \frac{(1+2i\sqrt{3}-3)+2}{3(1+2i\sqrt{3}-3)} \right) \\ &= 2\pi i \left(\frac{1}{4} + \frac{1}{12} (3 - i\sqrt{3}) \right) \\ &= \pi \left(i + \frac{1}{6} \sqrt{3} \right),\end{aligned}$$

while

$$\begin{aligned}\int_{\gamma} f(z) dz &= \int_0^{\pi} \frac{R^2 e^{2it} + 2}{R^3 e^{3it} + 8} \cdot i R e^{it} dt \\ &= i \int_0^{\pi} \frac{1 + 2e^{-2it}/R^3}{1 + 8e^{-3it}/R^3} dt \rightarrow i\pi \quad \text{as } R \rightarrow \infty.\end{aligned}$$

Now $\int_{\mathcal{C}} = \text{P.V.} \int_{-R}^R + \int_{\gamma}$, so

$$\text{P.V.} \int_{-\infty}^{\infty} f(z) dz = \pi \left(i + \frac{1}{6} \sqrt{3} \right) - i\pi = \frac{1}{6} \sqrt{3}.$$

11

5.(a) Put $z = \frac{n-1}{n}$. This means that $n = \frac{1}{1-z}$, so

$$\frac{1}{n+1} = \frac{1}{1/(1-z)+1} = \frac{1-z}{2-z}$$

The function $f(z) = \frac{1-z}{2-z}$ is analytic for all $z \neq 2$ and satisfies

$$f\left(\frac{n-1}{n}\right) = \frac{1}{n+1}$$

for all $n \in \mathbb{N}$.

(b) The sequence $\left(\frac{n-1}{n}\right)_{n=1}^{\infty}$ lies in D and converges to a limit which is also in D . The coincidence principle says that the function f is unique.

or identity theorem 2

6. First put $f(z) = -5z$ and $g(z) = z^4 + 3$. On the unit circle,

$$|g(z)| \leq 4 < 5 = |f(z)|,$$

so $f+g$ has exactly one zero inside this circle. Now let $f(z) = z^4$ and $g(z) = -5z + 3$. If $|z| = 2$, then

$$|g(z)| \leq 13 < 16 = |f(z)|,$$

so all the zeros of $f+g$ lie inside the circle $\{z \in \mathbb{C} : |z| = 2\}$. Thus $p(z)$ has at most 3 zeros in the annulus.

7